

YUHAN CHEN

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EDUCATION

Huazhong University of Science and Technology, China GPA: 4.42/5 (89.6/100) Rank: 18/313 (top5.7%)
B.E. in Computer Science and Technology Sept. 2023 – Jun. 2027 (Expected)

PUBLICATIONS

- [1] **Quantifying the Gap between Understanding and Generation within Unified Multimodal Models** [PDF]
Chenlong Wang* Yuhang Chen* Zhihan Hu* Dongping Chen, Wenhui Chen, Sarah Wiegrefe, Tianyi Zhou†
Accepted at **CVPR 2026 Findings**. *IEEE Conference on Computer Vision and Pattern Recognition*
- [2] **Judge Anything: MLLM as a Judge Across Any Modality** [PDF]
Shu Pu, Yaochen Wang, Dongping Chen, Yuhang Chen, Guohao Wang¹, Qi Qin, Zhongyi Zhang,
Zhiyuan Zhang, Zetong Zhou, Shuang Gong, Yi Gui, Yao Wan, Philip S. Yu,
Accepted at **KDD 2025 Datasets and Benchmarks Track**, *Knowledge Discovery and Data Mining*
- [3] **PAPER2WEB: LET’S MAKE YOUR PAPER ALIVE!** [PDF]
Yuhang Chen*, Tianpeng Lv*, Siyi Zhang, Yixiang Yin, Yao Wan†, Philip S. Yu, Dongping Chen†,
Accepted at **ACL 2026**, *Annual Meeting of the Association for Computational Linguistics*

IN SUBMISSION

- [1] **DIVE: Benchmarking Full-Duplex Interaction with Visual Engagement in Continuous Streams** [PDF]
Yuhang Chen*, Guangyue Liu*, Shilong Liu†
A benchmark for real-time full-duplex multimodal interaction, featuring 525 audio-visual samples across six tasks
spanning proactive intervention, continuous engagement, and interaction coordination.
- [2] **MOSAIC: Multi-Policy Sampling and Collaborative Test-Time Adaptation for Vision-Language-Action Models** [PDF]
Yuhang Chen*, Guangyue Liu, Yinchun Tian, Jialin Liu
A collaborative test-time adaptation framework for VLA policies, improving 103/104 cross-policy evaluations (+3.03 pp
on average) and achieving +2.91 pp within-policy gains across four robotic manipulation benchmarks.

RESEARCH INTEREST

- **World Model:** Interactive World Models, Spatial Reasoning, 3D-Aware World Model
- **Multi Modality:** Unified Multimodal Models and their Applications in Embodied AI

RESEARCH EXPERIENCE

Research Intern in **Columbia University in the City of New York**

Working with Prof. Shilong Liu.

June. 2026 – Present

- Research on **interactive world models for embodied AI**, exploring how learned environment dynamics can support perception, reasoning, planning, and interaction in open-ended environments.

Research Intern at **InSpatio**

Research on World Models and Embodied Intelligence

June. 2026 – Present

- Contributed to **InSpatio-World 1.5** [Project], focusing on **training-data preprocessing** and **training of a spatiotemporal generative architecture** for camera-controllable novel-view generation.
- Research on **World Action Models (WAMs)** for embodied intelligence, exploring the integration of world dynamics modeling and action generation.

Research Intern in **University of Maryland**

Working with Prof. Tianyi Zhou.

Sept. 2025 - Jan. 2026

- Research on **Token Efficient and Interpretable Reasoning**. We explore the inherent thinking pattern of reasoning models and the effectiveness of thinking tokens.
- Explore the **User-centric AI** and LLM capabilities of providing more options and inspiration to users.

- Research on **the gap between understanding and generation within unified multimodal models**. We propose an evaluation framework and a quantifying metric to investigate such a gap, and conduct in-depth empirical study to analyze the underlying mechanism.

Research Intern at **ONE Lab** in Huazhong University of Science and Technology

Working with *Prof. Yao Wan*.

Oct. 2024 - Feb. 2026

- Research on **Academic front-end code Generation**. We propose the first benchmark Paper2Web and an autonomous system PWAgent that utilizes MCP-driven tools to transform static PDFs into interactive, layout-aware project sites.
- Research on **Multimodal Consistency Evaluation**. We introduce a symmetric framework Judge Anything to evaluate MLLM-as-a-Judge across 15 any-to-any modality categories.
- Research on **Multimodal Agents**. We explore the integration of MLLMs with autonomous agentic workflows, leveraging multi-step reasoning and tool-use capabilities to enhance model performance in complex, open-ended cross-modal tasks.

PROJECTS

Nano World Model : A world-model-based framework, realizing the fine-grained instance manipulation within some causal spatial reasoning scenarios.

Paper2Web: Develop PWAgent, an autonomous system utilizing MCP-driven tools and iterative refinement to transform static PDFs into interactive, layout-aware academic homepages. [\[Code Repository\]](#)

Judge Anything: Propose TaskAnything and JudgeAnything benchmarks to evaluate MLLM-as-a-Judge across 15 any-to-any modality categories; develop OmniArena, an automated platform to mitigate evaluation biases and hallucinations in MMG tasks. (*Dataset and Evaluation Platform of 2025 KDD Paper*)

GapEval: Propose a bidirectional benchmark to quantify the cognitive coherence between understanding and generation in Unified Multimodal Models (UMMs), revealing the disjointed nature of cross-modal knowledge through empirical studies on knowledge manipulation. (*Dataset of 2026 CVPR Paper*)

HONORS & AWARDS

Academic Scholarships: Awarded **Outstanding Academic Performance Scholarship** (3 times) and **Technology Innovation Scholarship** (2 times) 2023 – 2025

Computer System Development Capability Competition: Won the **Third Prize** in the national final, focusing on hardware-software co-design and system-level optimization. 2025.07

RuiKang Robot Developer Competition: Awarded the **Second Prize** in the national developer challenge for innovative robotic control and perception algorithms. 2025.09

Global Campus Artificial Intelligence Algorithm Elite Competition: Won **First Prize** at the National Finals in the Algorithm Application Track of the 7th Global Campus Artificial Intelligence Algorithm Elite Competition 2025.12